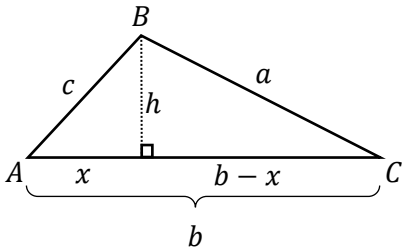
The Law of Cosines

Derivation

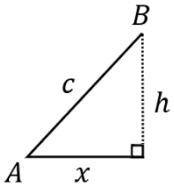
The Law of Cosines can be developed by combining two familiar topics, the ***Pythagorean theorem*** and ***right triangle trigonometry***.

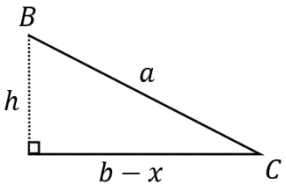
EXAMPLE 1

Complete each step outlined below to derive the rule for the Law of Cosines.



(a) Establish the necessary equations.





(eqn. 1) $\cos(A) = \frac{x}{c}$

(eqn. 2) $h^2 = c^2 - x^2$

(eqn. 3) $a^2 = (b-x)^2 + h^2$

$a^2 = b^2 - 2bx + x^2 + h^2$

(b) Substitute *(eqn. 2)* into *(eqn. 3)* and simplify.

$a^2 = b^2 - 2bx + \cancel{x^2} + (c^2 - \cancel{x^2})$

$a^2 = b^2 - 2bx + c^2$

(c) Solve *(eqn. 1)* for x, substitute this expression into your result from (b) and simplify.

$\cos(A) = \frac{x}{c}$

$x = c\cos(A)$

$a^2 = b^2 - 2b(c\cos(A)) + c^2$

$a^2 = b^2 + c^2 - 2bc\cos(A)$

Applying the Law of Cosines

THEOREM: ***Law of Cosines***

For any triangle ABC, with side measures a, b, and c, we have the following relationships.

$$a^2 = b^2 + c^2 - 2bc\cos(A)$$
$$b^2 = a^2 + c^2 - 2ac\cos(B)$$
$$c^2 = a^2 + b^2 - 2ab\cos(C)$$

EXAMPLE 2

Solve the triangle ABC given $A = 62.5°$, $b = 12$ cm, and $c = 17$ cm.

We know

$A = 62.5$
$b = 12$
$c = 17$

We need

$B =$
$C =$
$a =$

Find a

$a^2 = b^2 + c^2 - 2bc\cos A$
$a^2 = (12)^2 + (17)^2 - 2(12)(17)\cos(62.5)$
$a^2 \approx 244.60656658$
$a \approx 15.64$ $(15.639903\ldots)$

Find B

$b^2 = a^2 + c^2 - 2ac\cos(B)$
$\cos(B) = \dfrac{b^2 - a^2 - c^2}{-2ac}$
$\cos(B) = \dfrac{(12)^2 - (15.639903)^2 - (17)^2}{-2(15.639903)(17)}$

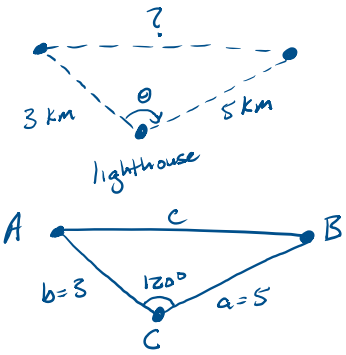
$\cos(B) \approx 0.732678242$
$\approx \cos^{-1}(0.732678242)$
$\approx 42.89°$ $(42.888660704\ldots)$

Find C

$A + B + C = 180$
$C \approx 180 - 62.5 - 42.888660704$
$C \approx 74.61$ $(74.611397296\ldots)$

EXAMPLE 3

Two ships are anchored near a lighthouse. One is located 5 km from the lighthouse while the other is 3 km away. The lighthouse beacon shines its light on one ship and then the other exactly 20 seconds later. If the beacon rotates at a rate of one revolution per minute, how far apart are these two ships?



$\frac{1 \text{ rev}}{\text{min}} = \frac{360°}{\text{min}} \rightarrow$ 120° rotation in 20 sec. $\rightarrow \theta = 120°$

We know

$C = 120°$

$a = 5$

$b = 3$

We need

$c =$

$c^2 = a^2 + b^2 - 2ab\cos(C)$

$c^2 = 5^2 + 3^2 - 2(5)(3)\cos(120°)$

$c^2 = 49$

$c = 7 \rightarrow$ The ships are 7 km apart.

THEOREM: *The Triangle Inequality (for a Proper Triangle)*

In a proper triangle, the sum of the lengths of any two sides must be greater than the length of the third remaining side. That is, if a, b, and c are any three sides of a triangle, then $a + b > c$.

EXAMPLE 4

Determine if constructing a triangle is possible with the given side measures. If so, solve the resulting triangle.

(a) 6 in, 3 in, 2 in

$6+3>2$ ✓ $6+2>3$ ✓ $2+3>6$ ✗ no such triangle exists

a b c

(b) 6 in, 3 in, 5 in

$6+3>5$ ✓ $6+5>3$ ✓ $3+5>6$ ✓

A triangle is possible

$\cos(A) = \frac{a^2-b^2-c^2}{-2bc} = \frac{6^2-3^2-5^2}{-2(3)(5)} = \frac{2}{-30}$

$\cos(A) = -\frac{1}{15} \rightarrow A = \cos^{-1}\left(-\frac{1}{15}\right) \approx 93.82°$

$\cos(B) = \frac{b^2-a^2-c^2}{-2ac} = \frac{3^2-6^2-5^2}{-2(6)(5)} = \frac{-52}{-60}$

$B = \cos^{-1}\left(\frac{13}{15}\right) \approx 29.93°$

$\cos(C) = \frac{c^2-a^2-b^2}{-2ab} = \frac{5^2-6^2-3^2}{-2(6)(3)} = \frac{-20}{-36}$

$C = \cos^{-1}\left(\frac{5}{9}\right) \approx 56.25°$

Check: $93.82 + 29.93 + 56.25 = 180$ ✓

Heron's Formula for the Area of a Triangle

THEOREM: *Heron's Formula*

For any triangle ABC, with side measures a, b, c, and semiperimeter

$$s = \frac{1}{2}(a + b + c),$$

the area of the triangle is given by the following formula.

$$Area = \sqrt{s(s-a)(s-b)(s-c)}$$

EXAMPLE 5

Revisit Example 4 and find the area of the resulting triangle.

$a = 6\text{in}$, $b = 3\text{in}$, $c = 5\text{in}$

$s = \frac{1}{2}(6+3+5) = 7$

$Area = \sqrt{7(7-6)(7-3)(7-5)}$

$= \sqrt{7(1)(4)(2)}$

$= \sqrt{56}\ \text{in}^2 \rightarrow Area \approx 7.48\ \text{in}^2$

EXAMPLE 6

A triangular-shaped field is bordered on two sides by a fence and on the third side by a relatively straight river. If the length along the river is 1257 ft and the fencing length is 1020 ft along one side and 1015 ft along the other, determine the acreage of the field. Round your answer to the nearest tenth of an acre (1 acre = 43,560 square feet).

$a = 1257\text{ft}$
$b = 1020\text{ft}$
$c = 1015\text{ft}$

$s = \frac{1}{2}(1257 + 1020 + 1015) = 1646$

$Area = \sqrt{1646(1646-1257)(1646-1020)(1646-1015)}$

$= \sqrt{1646(389)(626)(631)}$

$\approx 502911.495\ldots\ \text{ft}^2$

$Acreage \approx \frac{502911.495}{43560} \approx 11.5\ \text{acres}$